

ABINIT migration(s)

X. Gonze
UCLouvain, Belgium

ABInfrastructure for users & developers

User point of view :

- File server (abinitXXX.tar.gz)
- Web site (portal, documentation)
- Forum
- Mailing list
- Additional public repositories (e.g. pseudos)

Developer point of view :

- Gitlab server (ABINIT gitlab ... “git clone” from, “git push” to ...)
- Test farm (access login for debugging)
- Portal to the test farm (access login, also in the list of branches)
- Mailing list(s)

Who developed/maintains this infrastructure ?

ABInfrastructure for users & developers

User point of view :

- File server (abinitXXX.tar.gz)
- Web site (portal, documentation)
- Forum
- Mailing list
- Additional public repositories (e.g. pseudos)

Developer point of view :

- Gitlab server (ABINIT gitlab ... “git clone” from, “git push” to ...)
- Test farm (access login for debugging)
- Portal to the test farm (access login, also in the list of branches)
- Mailing list(s)

Who developed/maintains this infrastructure ?

=>2023, developed+maintained by Jean-Michel Beuken **only**.

In 2023, we realized that JM would not be replaced when leaving for pension, in **August 2026 !**



How to survive without Jean-Michel ?

JMB is so unique ! We love him ...

Concerted action in the framework of ABILOGIX

ABILOGIX : A consortium to manage the development of ABINIT.

Consortium agreement : 1/10/2021 – 30/9/2031.

Institutional value !

Members : UCLouvain, ULiège, UNamur, CEA (1/10/23), UTQR (1/9/25).

In practice : meetings for coordination of the actions of permanent members incl. Jean-Michel Beuken, Matteo Giantomassi, HeXu.

In 2023 : start of brainstorming + decisions + actions

Goals :

- Remove all single point failures/knowledge linked to JMB
- Make the infrastructure less human-time consuming to maintain
- Spread the work on more actors

=> better future for ABINIT : more resilient + maintainable !



2023 - 2026

Institutions and actors more directly involved : monthly meetings of **ABITECH**

He Xu + Philippe Ghosez for **ULiege**

Marc Torrent + Fabien Brieuc for **CEA**

Matteo Giantomassi + XG for **UCLouvain**

Decisions (also in concertation with core developers at biannual meetings)

1. Test farm physical machines to be migrated to HPC **UCLouvain CISM**
2. Virtual machines for “light” services rationalized, under **UCLouvain SGSI**
(forum, file server, pseudodojo, buildbot, BBportal ...)
3. Admin rights shared on these machines
4. DNS for abinit.org hosted by **UCLouvain** (Samuel Poncé+XG)
5. Migration of the UCLouvain Gitlab to **ULiège** institutional Gitlab (He Xu)
6. Adaptation of buildbot+BBportal infrastructure and hosting
7. Continuation of hosting mailing lists at **UCLouvain** (sympa system)
8. Everything else on **Github**
9. Maintenance of ABINIT software stack of testfarm **CEA** (Marc Torrent+CEA)
10. Forum user management by **UTQR** (Gabriel Antonius)



2023 – 2026 actions : what is done ?

Decisions (also in concertation with core developers at biannual meetings)

1. Test farm physical machines to be migrated to HPC **UCLouvain CISM**
SOON : to be described now (JMB+CISM+Xavier)
2. Virtual machines for “light” services at **UCLouvain SGSI**
(forum, file server, pseudodojo, buildbot, BBportal)
DONE (thanks to JM)
3. Admin rights shared on these machines
DONE (ABITECH + CISM + SGSI)
4. DNS for abinit.org hosted by **UCLouvain** (Samuel Poncé+XG) **OPERATIONAL**
5. Migration of the Gitlab to the **ULiège** institutional Gitlab
SOON : to be described now (He Xu)
6. Adaptation of buildbot+BBportal infrastructure and hosting **UCLouvain**
SOON : to be described now (Matteo Gantomassi)
7. Continuation of hosting mailing lists at **UCLouvain** (sympa system) **OK**
8. Everything else on **Github** **OK**
9. Maintenance of ABINIT software stack of testfarm **CEA** (Marc Torrent+CEA)
CEA (Marc Torrent) gained the needed expertise
10. Forum user management by **UTQR** (Gabriel Antonius) **OK**



2023 – 2026 : reliance on Github

Was already used before 2023 : [Abipy](#) !

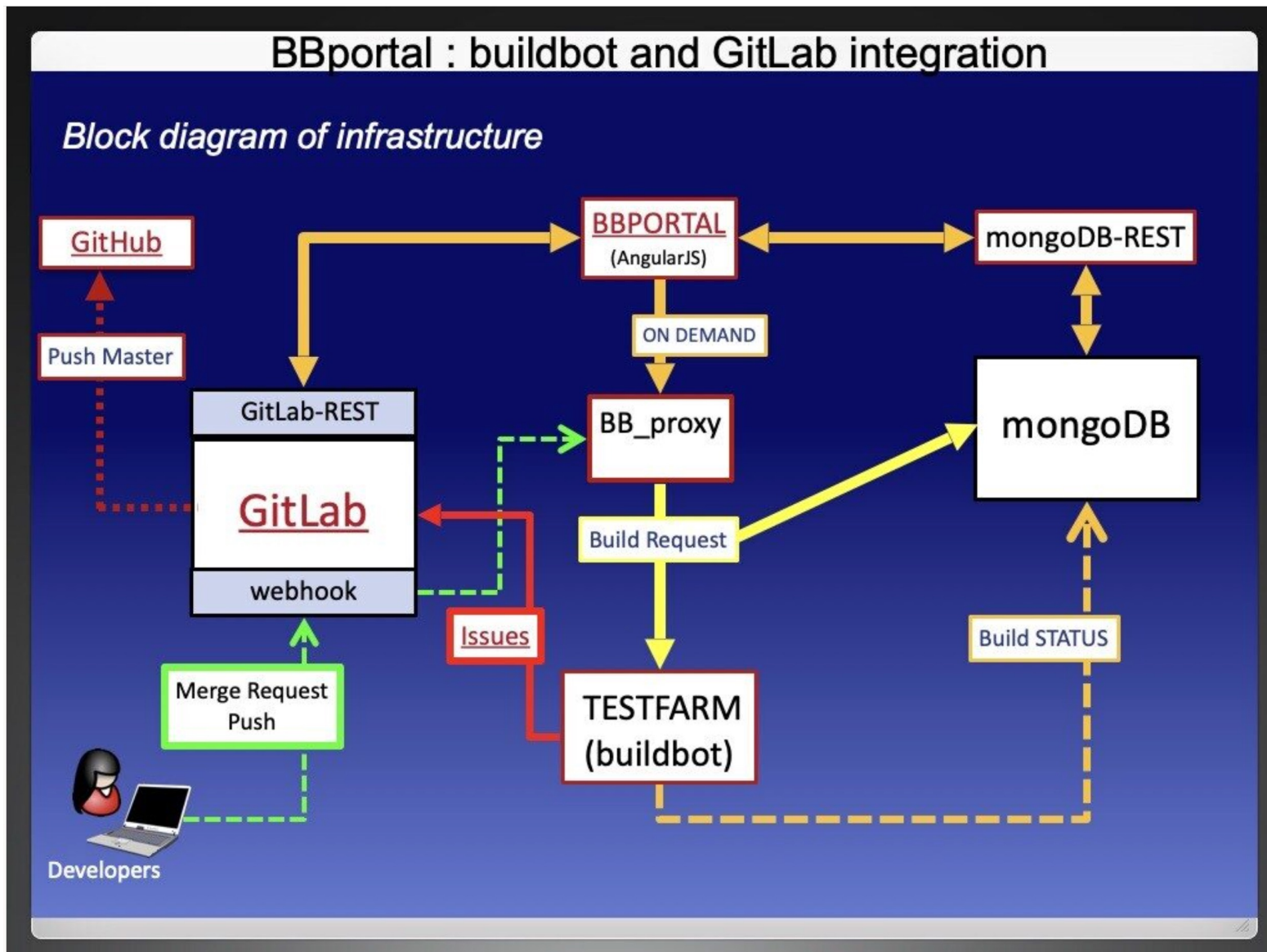
On Github now :

- Web site ([migration by M. Azizi](#))
- Delivery of ABINIT tarballs
- All pseudos delivered from pseudo-dojo Github
- Access to previous ABINIT versions
- Hosting : present conference site
- Hosting : big data files for some ABINIT tutorials, also benchmarks
- Hosting : the abimkdoc documentation, including previous versions
- Hosting : ABINIT private documents (ABILOGIX, procedures, etc)
- Hosting : scripts and data for abibuildbot
- ...

Current view on Gitlab/Buildbot/BBPortal

jmb_2023_abidev copy.key

Close



Current view BBPortal : summary


GitLab3

Buildbot Status
On Demand
Merge Requests
Logout


filters :  all  all  all 
night 
17:54:41

Summary of active branches info

Branch	rev	Date	ACO	LRS	AOE	NAG	NGS	alO	aTQ	BO7	EOF	ENA	EOZ	EIC	EOE	EOI	MANN	MON	SCO	S64	SCY	SCP	SCD
sarrautem/autotools-tests-and-misc-10.8/43fd4805 [NI]	10.8.2	27/05 17:17																					
trunk/release-10.8/21b18abd [NI]	10.8.2	26/05 23:50																					
trunk/develop/823e6d21 [OD]	10.9.0	24/05 22:26																					
trunk/develop/823e6d21 [NI]	10.9.0	24/05 21:43																					
xuhe/optim_dfpt_io/0029d117 [RF]	10.7.2	24/05																					

Current view BBPortal : on-demand

 GitLab3

Buildbot Status

On Demand

Merge Requests

On Demand Form

Submit

Start build... 

💬 Job submitted : trunk/release-10.8/21b18abd on night

Branch

User trunk 🔗 Branch release-10.8  </> Commit 21b18abd 💬 Comment : Fix typo in tbase3.md

Builder

🌐 Type night scope_gnu_10.2_parallel, scope_gnu_12.2_mpich, scope_gnu_12.2_abipy,
eos_nvhp_23.9_elpa, scope_gnu_13.2_dep, eos_gnu_13.2_mpich, eos_gnu_13.2_serial,
alps_gnu_9.5_sdebug, eos_nvhp_24.9_openmpi, alps_nag_7.2_serial,

Migration of the test farm to CISM

To be done physically in June/July/August

(1) ALPS then (2) SCOPE then (3) EOS

Others : left outside CISM, removed when some problem will occur...

Buildbot testfarm server matrix

The current reference [builder](#) is "eos_gnu_13.2_mpich", running on the ser

Name	CPU Type <i>GPU Type</i>	#Cores (#THD)	RAM (GB)	OS
alps	Xeon 6230	2x20 (80)	64	Rocky-9.5
atlas	Xeon E5-2623	2x8 (16)	64	Fedora-41
bob	Xeon E5-2603	2x6 (12)	16	Fedora-39
eos	AMD 7643 <i>2 * A30</i>	2x48 (192)	256	Ubuntu-22.04
minimac	Apple M1 Ultra	16+4	64	macOS-15.5
scope	AMD 7502	2x32 (128)	96	Ubuntu-18.04
manneback	Milan,EPYC,7763	40	2	Rocky 8

Purchase	S/N
6/20 3Yrs	HP DL360 G10 CZ20190LT4
6/15 3Yrs	Transtec CALLO
4/16 4Yrs	Dell R430 PowerEdge
12/22 4Yrs	Dell R7525 PowerEdge
2/23 3Yrs	Apple studio M1
6/20 3Yrs	HP DL385G10 CZJ520082V
	cluster keira

Then, money available
to buy new server

MANN : already
inside CISM !

Migration of the test farm to CISM

The MANN bot : a prototype ! On Keira partition of Manneback
Access has been recently granted to several developers, and documented

https://www.abinit.org/buildbot.html#accessing_directly_some_CISM_server

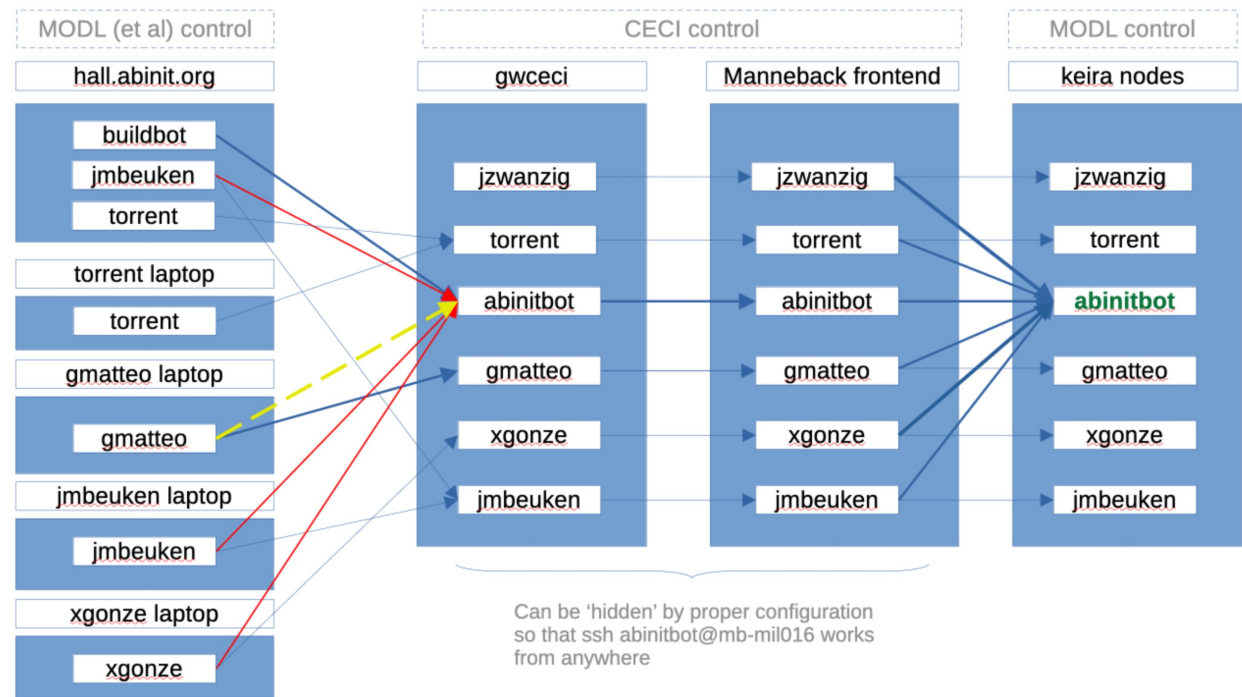
Particulars of the use of buildbot for ABINIT

Triggering the execution of the automatic tests

How to handle failures ?

Accessing directly some server in the MODL network.

Accessing directly some server in the CISM environment.



https://github.com/abinit/abinit_private_docs/blob/main/Abitools/abinitbot.png

The future : who will do what ? who will help ?

Several people from ABITECH should be able to perform the basic tasks (the documentation should be available or made available), but some might be more “usual” people ...

See slide 6.

But also : who will create accesses / new bots ... ?

Create access to Gitlab (when in ULiège) : [He Xu](#)

Create access to bbportal : [Matteo](#)

Create access to test farm when in CISM : [Xavier](#)

Create access to Abinit Github : [Xavier](#)

Create/Maintenance of new builders : [Marc+CEA](#)

Monitor/maintain the forum : [Gabriel](#)

Who will monitor the test farm machines : [He Xu](#)

Maintenance of (new) virtual machines : [Marc+CEA](#)